Pre-Analysis Plan

PAP guidelines
Please detail the exact way you will test each hypothesis. In particular, please consider the following for each hypothesis:
Variables
Which variables will you include? How will these variables be coded – will you use them the way they are or recode them, and if so, how?
Inclusion/exclusion criteria
Are there any observations you will exclude? If yes, which? Which years will you include/what time period will you study?
Exact analysis
Please write down the exact specification of the regression model, including the names of the dependent and independent variables and whether you use robust standard errors or something else.
Interpretation of results
Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test is in line or not in line with the hypothesis?

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).
That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters) > (X for respondents without daughters).

Variables

Dependent Variable:

plh0308_2 (best if man and woman work the same amount so they can share the responsibility, 7-point scale from do not agree at all to completely agree); renamed to gender_sharework and removed missing values

Independent Variables:

Treatment: daughter
Dummy indicator (daughter), where daughter=1 if respondent has at least one daughter that was born prior to the survey year; daughter=0 if respondent has only sons or no children at the time of survey

```
// ever has a daughter
        egen daughter_ever = rowmax(kidsex01-kidsex19)
        mvdecode daughter_ever, mv(-1)
        replace daughter_ever = daughter_ever-1

// has a daughter one year prior to survey year
        mvdecode kidgeb* pgmonth, mv(-3 -2 -1)
        gen has_daughter = 0
```

```
forvalues i = 1/19 {
local s = string(`i', "%02.0f")   // to match 01, 02, ..., 19
replace has_daughter = 1 if kidsex`s' == 2 & kidgeb`s' < syear & !missing(kidsex`s') &
!missing(kidgeb`s') & !missing(syear)
}

// indicator: has daughter vs. never has daughter (only son(s))
gen daughter = .
replace daughter = 0 if has_daughter==0 & daughter_ever==0
replace daughter = 1 if has_daughter==1 & daughter_ever==1
```

Control/matching variables

- Missing values
  ```
  mvdecode pgcasmin sex pglfs pgemplst pgfamstd, mv(-10/-1)
  mvdecode plh0004 plh0308_v2 pli0012_h-pli0044_h, mv(-8/-1)
  ```
- gender (male vs female)
- age (in years)
- pgcasmin
  ```
  recode pgcasmin (1=1)(2=2)(3=3)(4 5=4)(6=5)(7=6)(8=7)(9=8)(else=.), gen(edu)
  ```
- emplstat (employment status: full-time, part-time, vocational training/in education, retired, unemployed, other)
  ```
  gen emplstat = .
  replace emplstat = 1 if pgemplst==1 // full-time
  replace emplstat = 2 if pgemplst==2 | pgemplst==3 // part-time
  replace emplstat = 3 if pgemplst==5 & pglfs == 2 // nw: retired
  replace emplstat = 4 if pgemplst==5 & pglfs == 6 // nw: unemployed
  replace emplstat = 5 if pgemplst==5 & (pglfs==1 | pglfs == 3 | pglfs==5 | pglfs==8 |
  pglfs==9 | pglfs==10 | pglfs==13) // nw: other
  replace emplstat = 5 if pgemplst==3  // nw: other (vocational training)
  ```
- pgfamstd (as is)
- east/west
  ```
  recode hgnuts1_ew (21=0 "west")(22=1 "east")(else=.), gen(east)
  ```
- hhinc (net equivalent household income)
  ```
  recode hgtyp2hh (11 12 13=1)(21 31 33=2)(32 34 35 41 51=3)(36 42 52 61=4)(43 53
  62=5)(else=.), gen(hhsize) // approximation, as hhbrutto dataset is not available
  gen hhinc = hghinc/(sqrt(hhsize))
  ```
- Number of children:
  ```
  gen n_children = 0

  forvalues i = 1/19 {
    local s = string(`i', "%02.0f")
  ```

```
 replace n_children = n_children + 1 if kidgeb`s' < syear & !missing(kidgeb`s') &
!missing(syear)
}
recode n_children (5/30=5), gen(n_children6)        // recode to adjust for potential
outliers
```

Inclusion/exclusion criteria

Exclusion of respondents who never had children during the entire survey period (keep if sumkid>0)
Exclusion of respondents where partner lives abroad, and same-sex partnerships (keep if pgfamstd<6)
Exclusion of respondents who are still in school (keep if pgcasmin>0)
Exclusion of respondents below the age of 18 (gen age = syear - gebjahr -> keep if age>=18)

For H1: plh0308_v2 is only available in 2018

Exact analysis

Average treatment effect of the treated (ATET), where daughter=1 is the treatment, using propensity score matching to estimate the causal effect of having daughters.

teffects psmatch (gender_sharework) (daughter sex age i.edu i.emplstat i.pgfamstd n_children6 hhinc east, probit), atet

Interpretation of results

A positive, significant effect (ATET daughter 1 vs 0) on gender_sharework is in line with hypothesis

H2. Having daughters leads to more left-leaning political preferences.
That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters) > (Y for respondents without daughters).

Dependent variable:
Left-right self-placement ("Political attitude"), where 0=very left and 10=very right.
gen leftright = plh0004
removed missing values

Independent variables:
Identical to H1

Inclusion/exclusion criteria
Identical to H1

Exact analysis
Average treatment effect of the treated (ATET), where daughter=1 is the treatment, using propensity score matching to estimate the causal effect of having daughters.

teffects psmatch (leftright) (daughter sex age i.edu i.emplstat i.pgfamstd n_children6 hhinc east i.syear, probit), atet

Interpretation of results

A negative, significant effect (ATET daughter 1 vs 0) on leftright is in line with the hypothesis

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters) > (Z for respondents without daughters).

Dependent variable:

We use pli0043_h, pli0012_h, pli0016_h (household work) and pli0019_h, pli0022_h, pli0044_h (childcare work) as behavioral measures of gender equality (H3).
We calculated the average hours of household work and childcare work for each respondent and then calculated a mean index across both. Likewise, we calculated these for each of the respondents' partners. Afterwards, we calculated the hourly share that each respondent participates in the household (doing childcare or household work). For this share, we calculate the absolute deviation from 0.5 (as a state of equality). The higher the deviation, the more unequal the housework share. As we only look at heterosexual relationships, this indicates gender (in)equality.
Missing values of the initial variable were removed.

```
gen hh_work = (pli0043_h + pli0012_h + pli0016_h)/3
gen hh_childcare = (pli0019_h + pli0022_h + pli0044_h)/3
gen hh_workcare = (hh_work + hh_childcare)/2

gen p_hh_work = (p_pli0043_h + p_pli0012_h + p_pli0016_h)/3
gen p_hh_childcare = (p_pli0019_h + p_pli0022_h + p_pli0044_h)/3
gen p_hh_workcare = (p_hh_work + p_hh_childcare)/2

gen share_hh_workcare = hh_workcare/(hh_workcare+p_hh_workcare)
gen dev_share_hh_workcare= abs(0.5-share_hh_workcare)
```

Partner variables starting with "p_" are first matched to respondents.

Independent variables:

Identical to H1

Inclusion/exclusion criteria

Identical to H1

Exact analysis

Average treatment effect of the treated (ATET), where daughter=1 is the treatment, using propensity score matching to estimate the causal effect of having daughters.

```
teffects psmatch (dev_share_hh_workcare) (daughter sex age i.edu i.emplstat i.pgfamstd n_children6 hhinc east i.syear, probit), atet
```

Interpretation of results

A negative, significant effect (ATET daughter 1 vs 0) on dev_share_hh_workcare is in line with the hypothesis

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

That is, denoting X as a measure of support for gender equality,
(X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters).

Dependent variable:
Identical to H1

Independent variables:
Identical to H1

Inclusion/exclusion criteria
Identical to H1

Exact analysis
Average treatment effect of the treated (ATET), where daughter=1 is the treatment, using propensity score matching to estimate the causal effect of having daughters.

eststo m1: teffects psmatch (gender_sharework) (daughter age i.edu i.emplstat i.pgfamstd n_children6 hhinc east, probit) if female==0, atet
eststo m2: teffects psmatch (gender_sharework) (daughter age i.edu i.emplstat i.pgfamstd n_children6 hhinc east, probit) if female==1, atet

Then calculate the effect difference between both models:

Extract the required statistics

```
* (i) coefficient estimate
estimates replay m1
mat coef_men=r(table)
scalar N_men = e(N)

estimates replay m2
mat coef_women=r(table)
scalar N_women = e(N)

scalar atet_men = el(coef_men,1,1)
scalar atet_women = el(coef_women,1,1)

* (ii) standard error
scalar se_men = el(coef_men,2,1)
scalar se_women = el(coef_women,2,1)

* (iii) family of the test statistic (e.g. z, t, F, Chi²)
local test_stat_family = "z"
```

```
* (iv) residual and model degrees of freedom (DFs) if applicable: NOT APPLICABLE
local df_residual = e(df_r)
local df_model = e(df_m)

* (v) calculate p-value
// Calculate difference in ATETs and its standard error
scalar diff = atet_men - atet_women
scalar se_diff = sqrt(se_women^2 + se_men^2)

// Calculate z-statistic and p-value
scalar z = diff / se_diff
scalar p = 2 * (1 - normal(abs(z)))
local coefficient = diff
local se = se_diff
local p_value = p
```

Interpretation of results

A positive, significant effect (ATET daughter 1 vs 0) is in line with the hypothesis and indicates that the effect of having a daughter is stronger among men compared to women.